Example 3.2 Express the following numbers in exponential form with the given base:

(i) 1000, base 10 (ii) 512, base 2 (iii) 243, base 3.

2	512
2	256
2	128
2	64
2	32
2	16
2	8
2	4
	2

3	243
3	81
3	27
3	9
	3

Solution

(i) $1000 = 10 \times 10 \times 10 = 10^3$

(ii) $512 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^9$

(iii) $243 = 3 \times 3 \times 3 \times 3 \times 3 = 3^5$

Example 3.3 Find the value of (i) 13^2 (ii) $(-7)^2$ (iii) $(-4)^3$

Solution

(i) $13^2 = 13 \times 13 = 169$

(ii) $(-7)^2 = (-7) \times (-7) = 49$

(iii) $(-4)^3 = (-4) \times (-4) \times (-4)$
$= 16 \times (-4) = -64$

Note

$(-1)^n = -1$, if n is an odd natural number.

$(-1)^n = 1$, if n is an even natural number.

Think

Can you find two positive integers 'a' and 'b' such that $a^b = b^a$? $(a \neq b)$

Example 3.4 Find the value of $2^3 + 3^2$

Solution

$2^3 + 3^2 = (2 \times 2 \times 2) + (3 \times 3)$
$= 8 + 9 = 17$

Example 3.5 Which is greater 3^4 or 4^3?

Solution

$3^4 = 3 \times 3 \times 3 \times 3 = 81$

$4^3 = 4 \times 4 \times 4 = 64$

$81 > 64$ gives $3^4 > 4^3$

Therefore, 3^4 is greater.

Example 3.6 Expand a^3b^2 and a^2b^3. Are they equal?

Solution

$a^3b^2 = (a \times a \times a) \times (b \times b)$

$a^2b^3 = (a \times a) \times (b \times b \times b)$

Therefore, $a^3b^2 \neq a^2b^3$

DO YOU KNOW?

Ancient Tamilians have used lot of greater numbers in their day-to-day life. Refer to Kanakkathikaram written by a saint Karinayanar who lived in Tamilnadu during 10th century. More interestingly, they fixed a unique name for every big number. For example, 'Arpudham' means Ten crores, 'Padmam' means 10^{14}, 'Anantham' means 10^{29} and the word 'Avviyatham' denotes 10^{35}.

Pingalandai Nigandu Vaipaadu, an ancient Tamil Mathematics treatise, also confirms the usage of such big numbers. It is a book of multiplication table.

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